

Problems

Module 5
Logic

a) Consider the following statements:

- a) All people who are earning are happy
- b) All happy people smile
- c) Someone is earning

Perform the following tasks:

- i) Represent above statements in FOL
- ii) Convert each to CNF
- iii) Prove that someone is smiling using resolution technique. Draw the resolution tree

1. All earning people are happy

$$Earning(x) \rightarrow Happy(x)$$

2. All happy people smile

$$Happy(x) \rightarrow Smile(x)$$

3. Someone is earning

$$Earning(a)$$

(After Skolemization, assume a specific person **a**)

Step 1: Known Fact

Earning(a)

Step 2: Apply Rule 1

If $Earning(x) \rightarrow Happy(x)$

Since $Earning(a)$ is true,

Happy(a)

Step 3: Apply Rule 2

If $Happy(x) \rightarrow Smile(x)$

Since $Happy(a)$ is true,

Smile(a)

**Conclusion (Forward Chaining)**

$\exists x \text{ Smile}(x)$

✓ Someone is smiling.

👉 Backward chaining = Goal-driven reasoning

Start from the goal and work backwards.

Goal:

$$\exists x \text{ Smile}(x)$$

Assume we want to prove:

$$\text{Smile}(a)$$

Step 1: What rule gives Smile(x)?

From Rule 2:

$$\text{Happy}(x) \rightarrow \text{Smile}(x)$$

To prove Smile(a), we must prove:

$$\text{Happy}(a)$$

Step 2: What rule gives Happy(x)?

From Rule 1:

$$\text{Earning}(x) \rightarrow \text{Happy}(x)$$

To prove Happy(a), we must prove:

$$\text{Earning}(a)$$

Step 3: Check KB

We already have:

$$\text{Earning}(a)$$

✓ True

So,

$$\text{Earning}(a) \Rightarrow \text{Happy}(a) \Rightarrow \text{Smile}(a)$$

✓ **Conclusion (Backward Chaining)**

$$\exists x \text{ Smile}(x)$$

✓ Proven.

<https://www.ques10.com/p/13518/consider-the-following-axioms/>

Convert the following statements into predicate logic

1. All kings are persons.
2. Every city in Maharashtra has temple.
3. An Apple a day keeps doctor away.
4. Anything anyone eats and is not killed by is food.
5. Square of 3 is 9.

Consider the following facts:

1. Steve only likes easy courses.
2. Science courses are hard.
3. All the courses in the basket_weaving department are easy.
4. BK301 is a basket_weaving course.

Find by resolution that "What course would steve like?"

Consider the following facts:

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What actions would you take to prove "Some who are intelligent can't read" using propositional logic [10]

1. Whoever can read is literate.
2. Dolphins are not literate.
3. Some dolphins are intelligent.